Lab 4

Aim: To identify and document functional requirements for the project.

R1: User Authentication & Access Control

Description: The system shall allow registered users to securely authenticate and access the system based on assigned roles.

R1.1 Login

Description: The function allow a registered user to authenticate and access the system.

- **State:** User not authenticated
- **Input:** Username/Email, Password
- **Output:** Role-based dashboard or error message
- **Processing:** Validate credentials and identify user role (Student/ Faculty/ Administrator).

R1.1.1 Validate Credentials

- **Processing:** Match credentials with stored records.

R1.1.2 Identify User Role

- **Processing:** Redirect to Student / Faculty / Admin dashboard.

R1.2 Logout

- **State:** User is authenticated
- **Input:** Logout request
- **Output:** User session terminated
- **Processing:** Invalidate the active session.

R1.3 Password Management

- **Input:** Old password, New password
- **Output:** Password update confirmation
- **Processing:** Validate old password and securely update new password.

R2: User Profile Management

Description: The system shall allow authenticated users to view and update their own permitted profile details.

R2.1 View Profile

- **State:** User is authenticated
- **Input:** Profile view request
- **Output:** Display user profile details
- **Processing:** Retrieve profile information of the logged-in user.

R2.2 Update Profile

- **State:** User is authenticated
- **Input:** Editable profile details (Name, email, password, contact details)
- **Output:** Profile update confirmation
- **Processing:** Validate input data and Update user record

⚠ **Note:** *Role, course, account status and batch cannot be modified by the user.*

R3: User Account Administration

Description: The system shall allow administrators to manage all user accounts.

R3.1 Create User

Description: The function allows the administrator to create a new user account.

- **Input:** User details, role
- **Output:** Account creation confirmation
- **Processing:** Validate user details and store account information.

R3.2 Update User

Description: The function allows the administrator to modify user details.

- **Input:** Role, course, batch changes
- **Output:** Update confirmation
- **Processing:** Validate inputs and update user records

R3.3 Deactivate User

Description: The function allows the administrator to disable a user account.

- **State:** User account exists
- **Processing:** Disable user access without deleting data

R4: Academic Structure Management

Description: The system shall manage academic entities as courses, batches, and faculty-course-batch association required for assignment handling.

R4.1 Course Management

Description: The system shall allow administrators to create and update course information used for academic activities.

R4.1.1 Create Course

Description: This function allows the administrator to add a new course to the system.

- **State:** Administrator is authenticated
- **Input:** Course code, course name, semester, department
- **Output:** Course creation confirmation
- **Processing:** Validate course details and store them in the database.

R4.1.2 Update Course

Description: This function allows the administrator to modify existing course details.

- **State:** Course exists in the system
- **Input:** Modified course details
- **Output:** Course update confirmation
- **Processing:** Validate inputs and update course records.

R4.2 Batch Management

Description: The system allow administrators to manage student batches for academic organization.

R4.2.1 Create Batch

Description: This function allows the administrator to add a new academic batch.

- **State:** Administrator is authenticated
- **Input:** Batch name, academic year
- **Output:** Batch creation confirmation
- **Processing:** Store batch information in the system.

R4.2.2 Assign Students to Batch

Description: This function allows the administrator to assign student to an academic batch.

- **State:** Batch exists
- **Input:** Student list, batch identifier
- **Output:** Student-batch mapping confirmation
- **Processing:** Associate students with the selected batch.

R4.3 Faculty-Course-Batch Mapping

Description: The system shall allow administrators to assign a faculty member to a course and batch.

- **State:** Course and batch exist
- **Input:** Faculty ID, course, batch
- **Output:** Faculty assignment confirmation
- **Processing:** Store faculty-course-batch mapping details

R5: Assignment Management

Description: The system shall allow faculty to create, publish, and manage assignments for academic courses.

R5.1 Create Assignment

Description: The function allows faculty members to create a new assignment.

- **State:** Faculty is authenticated
- **Input:** Assignment title, description, start date, due date, maximum marks
- **Output:** Assignment creation confirmation
- **Processing:** Validate assignment details and store them in the system.

R5.2 Publish / Unpublish Assignment

Description: The function allows the faculty members to control the visibility of an assignment.

- **State:** Assignment exists
- **Input:** Visibility selection (Publish / Unpublish)
- **Output:** Assignment visibility update confirmation
- **Processing:** Update assignment availability for students.

R5.3 Assignment Browsing

Description: The function allows users to view assignments relevant to their enrolled courses.

- **State:** User is authenticated
- **Input:** Course selection
- **Output:** List of assignments for enrolled courses
- **Processing:** Retrieve assignments based on user enrolment and role.

R6: Assignment Submission Handling

Description: The system shall allow students to submit, revise, and track assignment submissions.

R6.1 Submit Assignment

Description: The function allows students to submit an assignment.

- **State:** Assignment is published and within deadline
- **Input:** Assignment file
- **Output:** Submission confirmation with timestamp
- **Processing:** Validate the submitted file and store the submission details.

R6.1.1 Validate File

Description: This function validates the submitted assignment file.

- **Processing:** Check file size and permitted file format.

R6.1.2 Store Submission

Description: This function stores the assignment submission.

- **Processing:** Save the assignment file and record the submission timestamp.

R6.2 Revise Submission

Description: The function allows a student to revise an existing submission before the deadline.

- **State:** Submission exists and deadline not passed
- **Input:** Updated assignment file
- **Output:** Updated submission confirmation
- **Processing:** Replace the previous submission with the new file.

R6.3 Track Submission Status

Description: The function allows a student to view the current status of an assignment submission.

- **State:** User is authenticated
- **Output:** Submission Status
- **Status:** Not Submitted / Submitted / Late / Evaluated
- **Processing:** Retrieve and display current submission status.

R7: Evaluation & Feedback Management

Description: The system shall allow Faculty to evaluate student submissions and provide feedback.

R7.1 Evaluate Assignment

Description: The function allows faculty members to evaluate submitted assignments.

- **State:** Student submission exists
- **Input:** Marks, remarks, corrected file
- **Output:** Evaluation saved confirmation
- **Processing:** Store marks, feedback, and corrected file for the submission.

R7.2 View Evaluation

Description: The function allows student to view evaluation results.

- **State:** Submission has been evaluated
- **Output:** Marks and feedback
- **Processing:** Retrieve and display evaluation details for the student.

R8: Deadline & Policy Control

Description: The system shall manage assignment deadlines and enforce late submission policies.

R8.1 Extend Deadline

Description: The function allows faculty to extend assignment deadlines for specific student or batches.

- **State:** Assignment exists
- **Input:** New deadline, selected student or batch
- **Output:** Deadline update confirmation
- **Processing:** Update assignment deadline based on specific criteria.

R8.2 Late Submission Handling

Description: The function enforces late submission rules for assignment.

- **State:** Assignment submission attempted
- **Processing:** Compare submission time with the deadline and apply late submission policies.

R9: Notification & Dashboard

Description: The system shall notify users of important academic events and present summarized academic information.

R9.1 Notifications

Description: This function generates notifications for users regarding academic activities.

- **Triggers:** New assignment creation, approaching deadline, evaluation release
- **Output:** System notification
- **Processing:** Generate and deliver notification to relevant users.

R9.2 Dashboard View

Description: The system shall display a summary of academic activities to the user.

- **State:** User is authenticated
- **Output:** Pending assignments, submission status, evaluation summary
- **Processing:** Aggregate and display user-specific academic data.